

STA304 Midterm Test (Lec0201)

Tuesday October 17, 2023

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Instructions:

- Total points: 20
- The test is 75 minutes long.
- There are 13 pages including this page.
- Questions begin on the next page.
- Only this exam book will be marked. Write your answers only in the space provided in this booklet. Any work on the Formula Sheet booklet will not be graded.
- Total questions: 13 (10 multiple choice, 3 short answer)
- Please ensure you **fill in your multiple Choice selections (Q1 - Q10) in the scantron on page 13**. Multiple choice answers NOT on the scantron will NOT be graded.
- Please use a calculator and do not round numbers in calculations until the very end, to avoid rounding error.
- In the multiple choice options, please select the closest answer (if the options are numeric) or the most appropriate answer (if the options are text).
- If you believe there to be a typo please still answer the question (see bullet point above), but you can list the question here and we will look into it after:

Multiple Choice Questions

Q1 (1 point)

Read the following three statements:

Statement 1: Snowball sampling is a probability-based sampling technique which is useful for hidden populations.

Statement 2: If a high school population has 200 male students and 200 female students, then handing out a survey to the first 50 male students and first 50 female students who enter the school is an example of quota sampling.

Statement 3: Simple random sampling is a probability-based sampling technique.

Select ONE of the options below, to best describe which of the above statements are TRUE.

- a. Only Statement 2 is correct.
- b. Only Statements 1 and 2 are correct.
- c. Only Statements 2 and 3 are correct.
- d. All three statements are correct.
- e. None of the three statements are correct.

Q2 (1 point)

A company of 20 employees wishes to take a sample of 4 of the employees. Read the following three statements:

Statement 1: If they take a systematic sample then the probability of selecting any particular sample is $\frac{1}{4}$.

Statement 2: If they use `sample(1:20, size=4, replace=FALSE)` to select their sample then the probability of selecting any particular sample is $\frac{1}{4845}$.

Statement 3: If they use `sample(1:20, size=4, replace=TRUE)` to select their sample then the probability of selecting any particular sample is $\frac{1}{160000}$.

Select ONE of the options below, to best describe which of the above statements are TRUE.

- a. Only Statement 2 is correct.
- b. Only Statements 1 and 2 are correct.
- c. Only Statements 2 and 3 are correct.
- d. All three statements are correct.
- e. None of the three statements are correct.

Q3 (1 point)

A local politician is planning to conduct a Simple Random Sample on voters in her region. She wishes to estimate the *total* distance (in km) all voters travel to vote on election day. She would like the total distance to be correct within 150km. Last election day there were $N = 3854$ voters, and the standard deviation of the distance traveled of each voter was 3.3km. Assume you want to ensure to calculate the total with at least 90% confidence, what sample size should be taken?

- a. 1481
- b. 1858
- c. 2907
- d. 3217
- e. 3354

Q4 (1 point)

Which ONE of the following is TRUE?

- a. When using cluster sampling we assume that clusters are dissimilar/heterogeneous.
- b. When using stratified sampling we are sampling in such a way that we will have representation from each strata in our sample.
- c. An example of a linear regression model is $y = \beta_0 + \beta_1 x$
- d. An example of a logistic regression model is $\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x + \epsilon$.
- e. A Frequentist framework uses a prior distribution to make inference on the parameter of interest, and a Bayesian framework does not use any priors.

Q5 (1 point)

The professor of the class surveyed the students in STA304 and stratified by lecture section. The class has $N = 500$ students in total, 270 of which are in the Afternoon lecture section. The professor is interested in estimating the mean time spent working on the Assignment. Use the following summary table to calculate the *lower bound* of the 95% CI for the overall mean hours spent working on the Assignment.

Lecture section	n	mean	variance
Morning	23	17.81	4.34
Afternoon	27	18.12	3.99

- a. 16.8849
- b. 17.4417
- c. 17.8310
- d. 17.9165
- e. 18.5132

Q6 (1 point)

Which ONE of the following is an example of a leading question?

- a. "Why is this course so bad?"
- b. "How often do you attend lecture and which section do you attend?"
- c. "What is your age?"
- d. "Are you currently enrolled in STA304?"
- e. All of the above are leading questions.

Data for Questions 7, 8, 9, and 10

Recall that the `UCLdata` from lecture contained the following variables:

- `admit`: (binary) 1 indicates that the student was admitted to graduate school, and 0 indicates they were not admitted.
- `gre`: (numeric) the students' recorded GRE score.
- `gpa`: (numeric) the students' recorded GPA score.
- `rank`: (categorical) Indicates whether the student attended a Rank 1, Rank 2, Rank 3, or Rank 4 university for their undergraduate studies.

Recall, this is what the data looked like:

```
glimpse(UCLdata)
```

```
## Rows: 400
## Columns: 4
## $ admit <int> 0, 1, 1, 1, 0, 1, 1, 0, 1, 0, 0, 0, 1, 0, 1, 0, 0, 0, 0, 1, 0, 1~
## $ gre    <int> 380, 660, 800, 640, 520, 760, 560, 400, 540, 700, 800, 440, 760, ~
## $ gpa    <dbl> 3.61, 3.67, 4.00, 3.19, 2.93, 3.00, 2.98, 3.08, 3.39, 3.92, 4.00~
## $ rank   <int> 3, 3, 1, 4, 4, 2, 1, 2, 3, 2, 4, 1, 1, 2, 1, 3, 4, 3, 2, 1, 3, 2~
```

```
head(UCLdata)
```

```
##   admit gre  gpa rank
## 1     0 380 3.61    3
## 2     1 660 3.67    3
## 3     1 800 4.00    1
## 4     1 640 3.19    4
## 5     0 520 2.93    4
## 6     1 760 3.00    2
```

Q7 (1 point)

If we wish to compare the sample size, admittance rate, as well as the mean GPA score, and standard deviation of the GPA score of applicants from Rank 1, Rank 2, and Rank 3 universities, which code chunk would be most appropriate?

a. The code chunk below:

```
UCLAdat %>%
  select(rank==1 & rank==2 & rank == 3) %>%
  group_by(rank) %>%
  mutate(n, mean(gpa), sd(gpa))
```

b. The code chunk below:

```
UCLAdat %>%
  select(rank==1, rank==2, rank == 3) %>%
  group_by(rank) %>%
  summarize(n, mean(gpa), sd(gpa))
```

c. The code chunk below:

```
UCLAdat %>%
  filter(rank==1 & rank==2 & rank == 3) %>%
  group_by(rank) %>%
  summarize(n, mean(gpa), sd(gpa))
```

d. The code chunk below:

```
UCLAdat %>%
  select(rank, gre) %>%
  filter(!rank==4) %>%
  group_by(rank) %>%
  summarize(n=n(), a=mean(admit), y=mean(gpa), s=sd(gpa))
```

e. The code chunk below:

```
UCLAdat %>%
  filter(rank==1 | rank==2 | rank ==3) %>%
  group_by(rank) %>%
  summarize(n=n(), a=mean(admit), y=mean(gpa), s=sd(gpa))
```

Q8 (1 point)

Assume that in 2023, there were $N=484$ undergraduate applications made to UCLA graduate programs. We wish to make inference on the ratio of GRE scores and the GPA scores of students. The following code was run to perform this analysis and the output is below.

```
N=484
n=nrow(UCLAdata)

my.design <- svydesign(id=~1,
                      data=UCLAdata,
                      fpc=rep(N, n))

svyratio(~gre, ~gpa, design=my.design)

## Ratio estimator: svyratio.survey.design2(~gre, ~gpa, design = my.design)
## Ratios=
##          gpa
## gre 173.3679
## SEs=
##          gpa
## gre 0.6686081
```

Which ONE of the following statements about the above scenario (code and output) is FALSE?

- a. In this example we are assuming that a simple random sampling approach was used, based on the input in `svydesign()`.
- b. We would need to test $H_0 : R = 0$ vs $H_A : R \neq 0$, to assess if GRE and GPA scores are significantly different from one another.
- c. We would need to test $H_0 : R = 1$ vs $H_A : R \neq 1$, to assess if GRE and GPA scores are significantly different from one another.
- d. In this example we can see that GRE score is estimated to be approximately 173.3679 times larger than the GPA score.
- e. In this example we can see that standard error of the ratio of GRE scores to GPA scores is 0.6686081.

Q9 (1 point)

The following code was used to estimate the GRE score of students based on their GPA score and the rank of their university.

```
UCLA_gre_model <- lm(gre ~ gpa + as.factor(rank), data=UCLAdata)
summary(UCLA_gre_model)$coefficients
```

##	Estimate	Std. Error	t value	Pr(> t)
## (Intercept)	206.986270	50.50776	4.0981081	5.057237e-05
## gpa	117.232423	14.08652	8.3223102	1.421978e-15
## as.factor(rank)2	-5.054814	16.16565	-0.3126885	7.546823e-01
## as.factor(rank)3	-34.556549	16.68155	-2.0715427	3.895676e-02
## as.factor(rank)4	-25.856189	18.89300	-1.3685591	1.719150e-01

Which ONE of the following statements are FALSE?

- a. The predicted GRE score of a student from a rank 2 university with a 3.2 GPA is 577.0752.
- b. We expect that students at a rank 1 university have a GRE score of about 25.856 points higher than students at a rank 4 university, on average.
- c. The model we are estimating is $y = \beta_0 + \beta_1 x_{gpa} + \beta_2 x_{rank2} + \beta_3 x_{rank3} + \beta_4 x_{rank4} + \epsilon$.
- d. The number 117.232 means that for every one unit increase in GPA we expect the GRE score of a student to increase by 117.232 points, on average (for students at universities of the same rank).
- e. We cannot calculate the predicted GRE score of a student at a Rank 1 university with a 3.5 GPA.

Q10 (1 point)

The code below was used to estimate the admittance rate of students, into graduate school, based on their GPA score and the rank of their undergraduate university.

```
UCLA_admit_model <- glm(admit ~ gpa + as.factor(rank),
                        data=UCLAdata, family="binomial")

summary(UCLA_admit_model)$coefficients
```

```
##              Estimate Std. Error  z value Pr(>|z|)
## (Intercept)   -3.4636185  1.1003145 -3.147844 1.644794e-03
## gpa            1.0520527  0.3101698  3.391860 6.941981e-04
## as.factor(rank)2 -0.6809725  0.3141442 -2.167707 3.018098e-02
## as.factor(rank)3 -1.3919078  0.3419004 -4.071091 4.679341e-05
## as.factor(rank)4 -1.5942508  0.4152054 -3.839667 1.232011e-04
```

Based on the output above, calculate the predicted probability of admittance for each of the two students listed below, and select the ONE statement below that is TRUE.

Student	GPA	rank
Sam	3.1	2
Emily	4.0	1

- a. Sam's predicted probability of admittance is 0.8832.
- b. Emily's predicted probability of admittance is 0.6780.
- c. Sam's predicted probability of admittance is 0.7075.
- d. Emily's predicted probability of admittance is 0.7448.
- e. We cannot calculate the predicted probability of admittance for either student.

Short Answer Question

Q11 (3 points total)

A company is interested in learning more about its customer database ($N = 238$). They have a spreadsheet of the population and they also have each column/variable stored as its own vector. The first six rows of the population data look like this:

```
## # A tibble: 6 x 3
##   Name      Email      Loyalty_Member
##   <chr>    <chr>    <chr>
## 1 Mary Smith    ms9912@umail.com    Yes
## 2 Mary Smith    mary_smith@coldmail.com No
## 3 Blue Mokowski bm1987@rmail.com    No
## 4 Katy Lee      k_lee@hmbc.ca       No
## 5 Cody Henderson cody_henderson@bb.org No
## 6 Maya Rudolph  maya@snl.com        Yes
```

The company assumes that the mean age of their customer base is about 30 years old, with variance of 30. *Without using the `sample()` function*, write one or two lines of code to simulate an `age` variable (which consists of whole number values) for the entire population.

Short Answer Question

Q12 (3 points total)

You are working on designing a survey to collect data on students' mental health. You are wondering if you should make all the questions in the survey required (i.e., not optional) in order to ensure there are no missing values in your data, and it is easier to analyze (and easier to make the results fully reproducible). Identify one right and one impact to consider in this scenario and in one or two sentences comment on the appropriateness of making the questions required or optional when designing the survey.

Impact:

Right:

Comments:

Short Answer Question

Q13 (4 points total)

Assume you have a simple random sample without replacement of $Y_1, Y_2, \dots, Y_n$ taken from the population of $\{u_1, u_2, \dots, u_N\}$. Recall from class:

$$E(Y_i) = \mu \qquad \text{Var}(Y_i) = \sigma^2$$

$$E(\bar{Y}) = \mu \qquad \text{Var}(\bar{Y}) = \frac{\sigma^2(N-n)}{n(N-1)}$$

Derive $E[(N-1)\bar{Y}^2]$. Show your work. Write your final answer as an expression of μ, σ, N and n .